ATRIA

HEALTHCARE APPOINTMENTS

SOFTWARE REQUIREMENTS SPECIFICATION

What Atria must do, and why.

Requirements for the Atria healthcare appointment booking and reminder system in Cameroon: patients, four staff roles and tenant administrators, on a Flutter app, a React web app and an AWS serverless backend. Every requirement traces to a project objective, a context requirement or a design decision stated in this document.

DOCUMENT	REVISION	DATE	STATUS
ATR-SRS-001	Rev B	17 September 2026	Issued for review

PROJECT	TEAM	STANDARD	REQUIREMENTS
DIGISOL Internship 2026	Bryan · Thanksgiving	ISO/IEC/IEEE 29148	160

DOCUMENT CONTROL

REV	DATE	CHANGE	AUTHOR
A	11 September 2026	First issue.	Bryan
B	17 September 2026	Three service lines, session queue, typed durations, procedure requests, intake and care context, fee ledger, ratings kept off prices and listings, profile completion and guardians, person and staff membership split with fairness rules, channels, four range views, region pack, asynchronous reliability, roles fixed at account creation. Rewritten to stand alone.	Bryan

PREPARED	REVIEWED	APPROVED
Project team	Internship supervisor	DIGISOL

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1 Introduction

1.1 Purpose

This specification states what the Atria system must do and how well it must do it. It is written for the team who will build Atria, the supervisor who will assess it, and any clinic that evaluates it before a pilot. Everything a reader needs to understand a requirement, including where it comes from, is inside this document.

1.2 Scope

Atria lets patients book, follow, reschedule and cancel outpatient visits, reminds them by SMS, and gives clinic staff and administrators the tools to run the day, including for patients who never install anything. It is designed for outpatient care in Cameroon, where general consultations are queued in sessions, specialists are booked by time, and procedures are sized by staff.

IN SCOPE	OUT OF SCOPE
Booking on three service lines, session queues, rescheduling and cancellation	Clinical records, diagnoses, prescriptions and results
SMS reminders, with voice fallback in Phase 3	Payments and settlement (entitlement is recorded, no money moves)
Email confirmations and cancellation notices	Video or telephone consultations
Intake answers as care context for one visit	Tenant onboarding commercial terms, branding and billing
Staff consoles for four roles, attendance, fee ledger and quality feedback	Integration with the national platform (prepared for, not built)

1.3 Definitions

TERM	MEANING
Tenant	An institution using Atria, such as a clinic group or health district. Every record belongs to exactly one tenant.
Clinic	A physical site within a tenant, where appointments take place.
Service line	General consultation, specialist consultation or procedure.
Appointment type	What is booked inside a line, with its own duration, buffer, notice, fee band and intake form.
Session	A published window of general consultation capacity with a number of rostered GPs.
Queue ticket	A patient's place in a session, with a server-assigned position and a live arrival window.
Grid unit	The calendar resolution set by the tenant, 10 minutes by default.
Slot lock	The record that reserves one grid unit of a clinician's time. Its existence prevents double booking.

TERM	MEANING
Care context	The intake answers a patient sends ahead of one visit, stored apart from the appointment.
Entitlement	The refund band an outcome gives the patient: NONE, PARTIAL or FULL.
Region pack	Country data a tenant chooses at onboarding: currency, phone rules, languages, timezone and formats.
Proxy recipient	A person, such as a parent or caregiver, who receives a patient's reminders.
Authority	Cameroon's Personal Data Protection Authority, created by Law No. 2024/017.
MoSCoW	Priority scale used here: Must, Should, Could.

1.4 Normative references

REF	DOCUMENT
LAW	Republic of Cameroon. Law No. 2024/017 of 23 December 2024 relating to personal data protection.
STD	ISO/IEC/IEEE 29148:2018 requirements engineering; WCAG 2.2; HL7 FHIR R4; OWASP ASVS 4.0.

1.5 How to read a requirement

Every requirement has an identifier, a statement that can be tested, a priority, a delivery phase, a trace to its source in section 3, and a verification method.

FIELD	VALUES
Priority	Must needed for the phase to be accepted · Should expected, may slip one phase with agreement · Could desirable
Phase	PH1 the core booking and reminder service · PH2 full product surfaces · PH3 pilot readiness, before real patient data
Trace	BR-nn project objective · CX-n context requirement · AD-nn design decision
Verify	Test · Demonstration · Inspection · Analysis

2 Overall description

2.1 Product perspective

Atria is a new, self-contained system. The context diagram shows everything outside it that exchanges data with it: patients and guardians, four staff roles, the Google identity provider, the SMS carrier and the email provider.

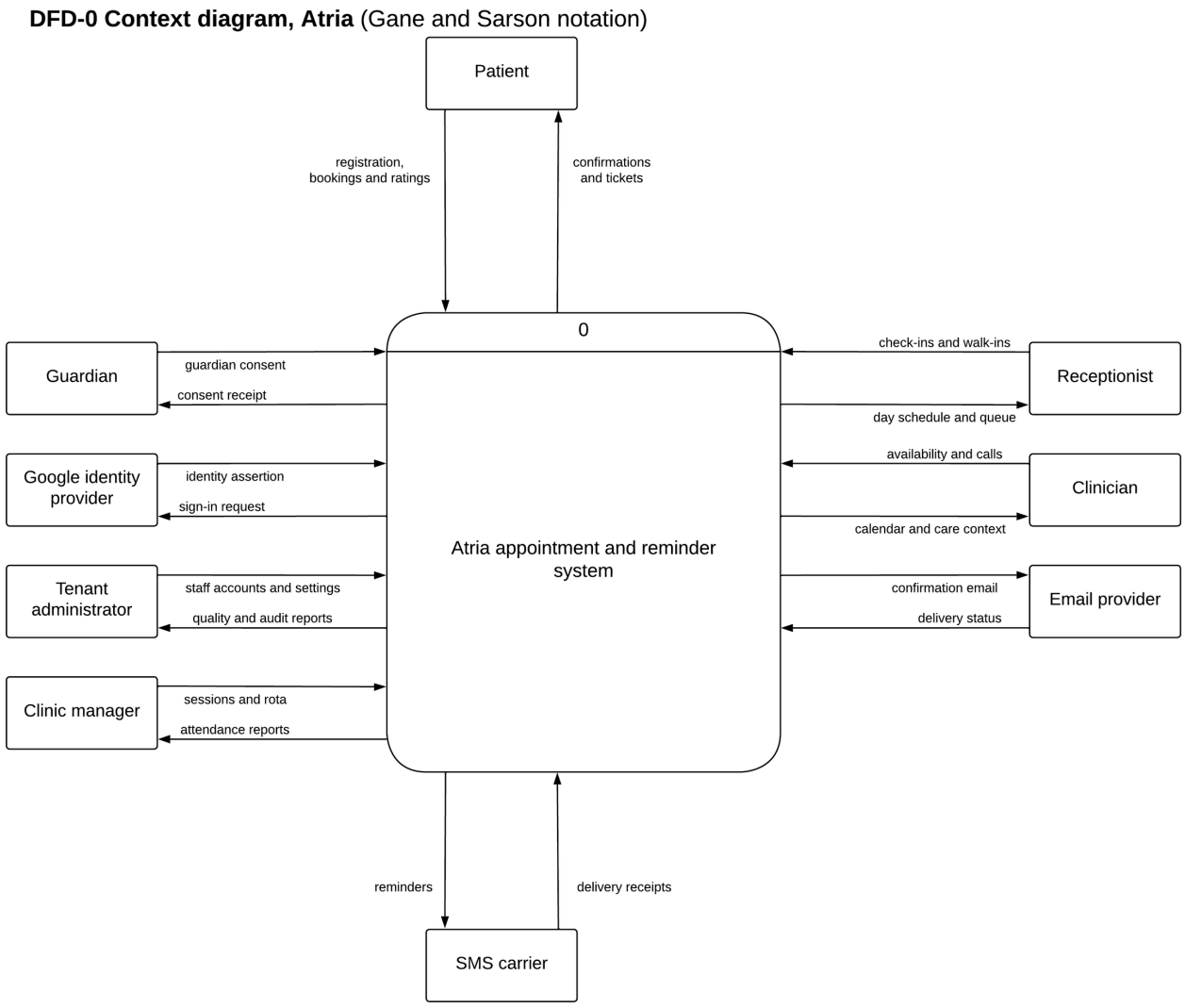


Figure 1. System context as a Level 0 data flow diagram in Gane and Sarson notation. Atria is one process; every external entity and the data it exchanges is shown.

2.2 User classes

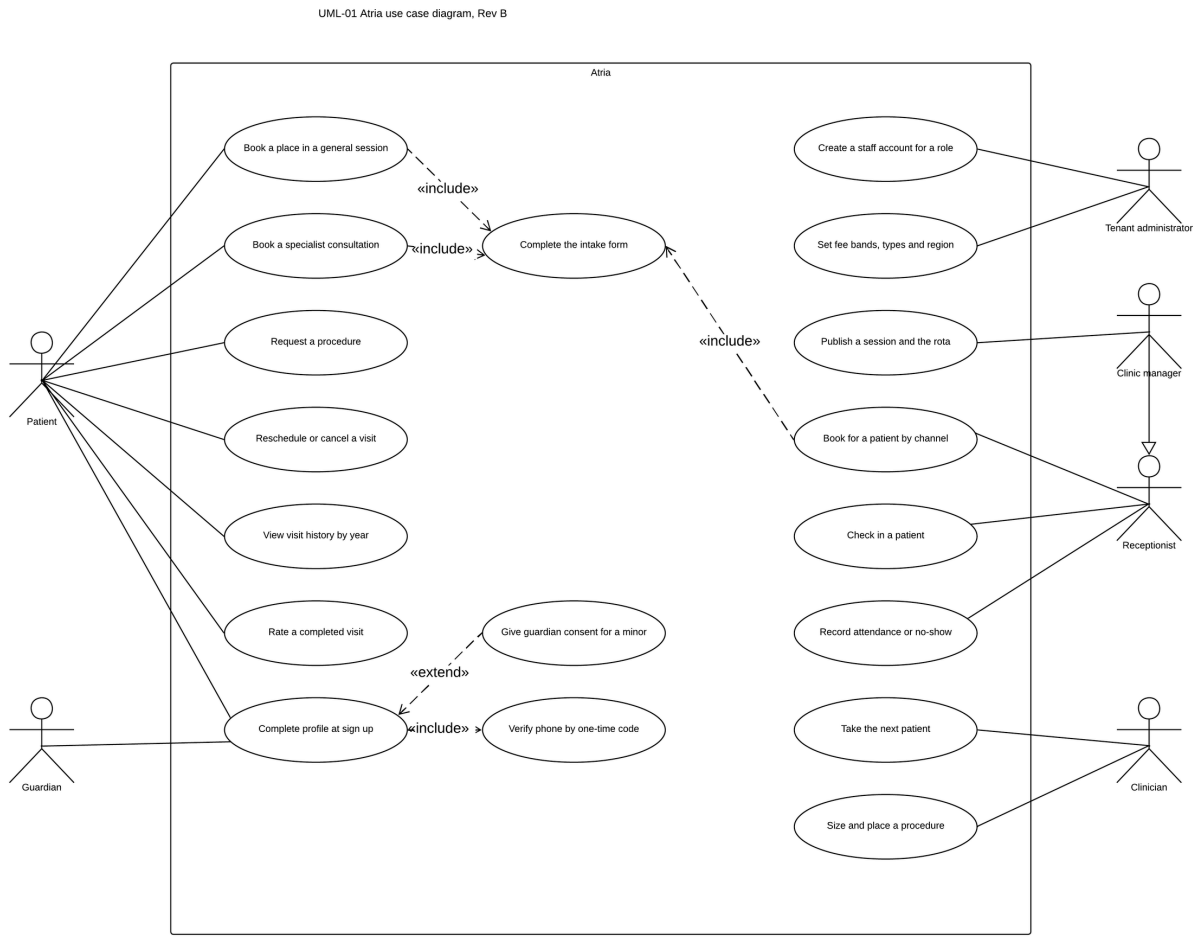


Figure 2. Use cases by actor.

CLASS	WHO	SURFACES	CHARACTERISTICS
Patient	Adults booking for themselves; guardians for minors	Flutter app (phone, tablet), React web (all sizes)	Varied literacy and connectivity. May never install the app; the SMS is the product for them.
Proxy recipient	Parent, caregiver or relative	SMS and voice only	Receives reminders; has no account.
Receptionist	Front desk	Flutter app (phone), React web (desktop, tablet)	Runs the day and the queue, books by phone and at the desk, never sees care context.
Clinician	Bookable health professional	Flutter app (phone), React web	Own calendar and availability, own patients' care context, own feedback.
Clinic manager	Runs one clinic	React web, Flutter app	Sessions and rota, reports, the weekly reschedule list, flagged reviews.

CLASS	WHO	SURFACES	CHARACTERISTICS
Tenant administrator	Practice manager or IT lead	React web, desktop only	Configures the institution, creates accounts with roles, reads audit and quality; never reads care context.
Scheduler	System actor	None	Fires reminders and feedback prompts at their time.

2.3 Operating environment

COMPONENT	ENVIRONMENT
Flutter app	Android 8.0 (API 26) and later; iOS 15 and later. Patient and staff roles; each account signs in to its own role.
React web app	Current and previous major versions of Chrome, Edge, Firefox and Safari, including mobile browsers. Patient and staff at every width; administration at 1,280 px and wider.
Backend	AWS: CloudFront, WAF, API Gateway (REST), Lambda, DynamoDB, S3, Cognito, EventBridge and Scheduler, SQS, SNS, SES, KMS, Secrets Manager, CloudWatch, X-Ray, CloudTrail, Budgets, Amplify Hosting.
Infrastructure	AWS CDK v2 in Python, one stack per concern.

2.4 Constraints

ID	CONSTRAINT	SOURCE
CON-01	The backend uses API Gateway, Lambda, DynamoDB, Cognito, SNS, SES and Secrets Manager.	BR-01 to BR-10
CON-02	Infrastructure is code, in AWS CDK.	BR-11
CON-03	No cloud provider has a region in Cameroon, so any region is a transfer of personal data abroad and needs the Authority's prior authorisation. Development uses synthetic data only.	CX-14
CON-04	Health data is sensitive data under Law No. 2024/017; processing needs the Authority's prior authorisation.	CX-13
CON-05	SMS is the primary reminder channel.	CX-4
CON-06	Fees follow a ministerial schedule and publicity about named doctors is prohibited, so ratings may not change prices or appear on a listing.	AD-11
CON-07	Most facilities do not have two doctors on site, so general consultations may not promise a named doctor at a minute.	AD-02

2.5 Delivery phases

PHASE	GOAL	EXIT CONDITION	REQUIREMENTS
PH1	The core service end to end: book, view and cancel, SMS reminders, email, role separation, on both clients.	Every objective's acceptance test in section 8 passes.	53

PHASE	GOAL	EXIT CONDITION	REQUIREMENTS
PH2	The full product: service lines, queues, consoles for four roles, administration, measurement and the fee ledger, bilingual interfaces.	Demonstrated to a clinic with synthetic data.	90
PH3	Pilot readiness: data protection compliance, voice fallback, offline staff writes, interoperability, payments.	The Authority's authorisations are held; no real patient data before this.	17

2.6 Assumptions and dependencies

- Each patient has a mobile number that can receive SMS, their own, a guardian's or a proxy's.
- SMS through SNS reaches MTN, Orange and Camtel with a registered sender ID. This is unverified; see OI-04.
- The pilot is hypothetical for the internship; compliance gates are documented and demonstrated rather than enforced.
- A pilot clinic would supply real hours, clinicians, fee bands and a named data protection contact before Phase 3.

3 Requirement sources

Requirements trace to three kinds of source, all stated here in full.

3.1 Project objectives

The project's objectives, evaluation metrics and deliverable, restated as business requirements.

ID	BUSINESS REQUIREMENT	KIND
BR-01	Provide a REST API through which patients book, view and cancel appointments.	Objective 1
BR-02	Authenticate patients and staff separately using Cognito.	Objective 2
BR-03	Send appointment reminders through Amazon SNS.	Objective 3
BR-04	Send confirmation and cancellation emails through Amazon SES.	Objective 3
BR-05	Store third-party service keys, such as an SMS or verification API key, in AWS Secrets Manager.	Objective 4
BR-06	Booking, retrieval and cancellation endpoints work end to end.	Metric 1
BR-07	Cognito distinguishes patient and staff access correctly.	Metric 2
BR-08	The SNS reminder fires ahead of the scheduled appointment time.	Metric 3
BR-09	SES sends confirmation and cancellation emails reliably.	Metric 4
BR-10	Secrets Manager holds every sensitive key; nothing is hardcoded in environment configuration.	Metric 5
BR-11	The system is deployed through infrastructure as code, with a report and an architecture diagram.	Deliverable

3.2 Context requirements

Requirements the Cameroonian setting imposes, with the reason each exists.

ID	CONTEXT REQUIREMENT	WHY
CX-1	Voice call fallback for reminders, selectable per patient	Many patients cannot read an SMS reliably or share a phone.
CX-2	Local-language message templates, configurable per tenant	French and English are official; local languages reach more patients.
CX-3	A nominated proxy recipient on the patient record	Children, older patients and shared phones need someone else to receive reminders.
CX-4	SMS-first delivery, with the web and app channels optional	Mobile subscriptions (108.2 per 100 people in 2024) far exceed internet use (46.3 per cent).
CX-5	Reminder lead times configurable per tenant	Travel times and clinic practice differ; no single lead time fits.
CX-6	Reminder count capped at two by default	More reminders add cost without evidence of benefit.

ID	CONTEXT REQUIREMENT	WHY
CX-7	Per-clinic baseline non-attendance measured from first deployment	No published Cameroonian outpatient non-attendance rate exists to compare against.
CX-8	Delivery status recorded per message	A reminder that did not arrive must be visible so the desk can phone.
CX-9	Tenant boundary modelled on the institution or health district	Care is organised by district and institution, not by single clinic.
CX-10	Staff-side scheduling usable without patient-side adoption	Many patients will book by phone or at the desk.
CX-11	Offline-tolerant staff client with queued writes	Coverage gaps are the most reported failure of existing SMS systems.
CX-12	No currency-denominated savings claim in product material	No local evidence supports a savings figure.
CX-13	Compliance with Law No. 2024/017 on personal data protection	Health data is sensitive data under the law, and the transition period ended in June 2026.
CX-14	Data Protection Authority authorisation, including for hosting abroad	No cloud provider has a region in Cameroon, so any hosting is a transfer abroad.
CX-15	Data modelled for exchange as HL7 FHIR resources	The national digital health plan funds an interoperability platform on FHIR.

3.3 Design decisions

Fifteen decisions settle how the product meets the objectives in this setting. Three follow directly from findings that contradicted the original plan: the emergency doctor is the scarcest person on site (AD-02, AD-03), fees and doctor listings are regulated (AD-11), and any hosting region is a transfer of data abroad (AD-13, NFR-CMP-04).

ID	DECISION	WHAT IT MEANS
AD-01	Three service lines	General consultation, specialist consultation and procedure are booked differently. Check-up and follow-up are appointment types inside a line, not lines of their own.
AD-02	Book a place in a session	The general line books a queue ticket in a published session. The patient gets a position and an arrival window, and the doctor is assigned at check-in by whoever is free.
AD-03	Disruption offers, it does not impose	When a clinician is marked unavailable, affected specialist appointments move to DISRUPTED and the patient is offered the next matching slots with one tap to confirm. Opt-in auto-accept applies to same-day slots only.
AD-04	Grid unit plus typed duration	The tenant sets a calendar grid unit (default 10 minutes) and every appointment type carries its own duration and buffer in grid units. A booking locks N consecutive units in one transaction.
AD-05	Procedures are staff-sized	A patient can request a procedure, which creates a REQUESTED appointment with no time held. Staff size and place it, and the patient confirms.

ID	DECISION	WHAT IT MEANS
AD-06	Intake per appointment type	Each type carries its own intake form. Answers are stored as care context: a separate entity, encrypted under its own key, readable only by the patient and the assigned clinician, and purged sooner than booking history.
AD-07	Fee ledger without moving money	Every appointment writes a ledger entry at closure with fee band, amount in XAF, outcome and refund entitlement. Nothing settles until a payment provider is added. No-show strikes are tenant-configurable and off by default.
AD-08	Four range views and a history horizon	Day, week, month (density grid) and year (twelve-month heatmap). Records are retained ten years and shown to patients five years by default, subject to confirmation (OI-12).
AD-09	Date of birth and a verified phone	Every patient confirms a Complete your profile step whatever the identity provider returns. Date of birth is stored, age is derived. The phone is verified by one-time code. Under 18 requires a linked guardian with recorded consent.
AD-10	One person, separate profiles, fair queues	A person holds a patient profile and, separately, staff memberships. Staff scope is dropped when the actor is the subject. Queue order is assigned by the server. Self-benefiting reschedules are blocked, every staff reschedule is reasoned and audited, and staff priority exists only as a granted quota.
AD-11	Ratings stay off prices and off names	Feedback is collected after attended visits and seen by the institution, never shown on a patient-facing clinician listing. Fees have a per-practice baseline. The ratingAffectsFee flag exists and is refused by the Cameroon region pack.
AD-12	Channel and type are two axes	Every booking records how it reached the clinic (online, walk-in, phone, referral) separately from what is booked. Referral opens referring facility and clinician fields and restricts the types to specialist and procedure.
AD-13	Region pack	Currency, phone rules, languages, timezone, holidays, address and ID formats, SMS sender policy, emergency number and default fees come from a region pack chosen at onboarding and locked after the first appointment. Cameroon: XAF, +237, French and English, Africa/Douala.
AD-14	Queues, dead letters, alarms and a budget	Every asynchronous consumer reads from SQS with a dead letter queue. CloudWatch alarms, X-Ray on the booking path, AWS Budgets with anomaly detection, CloudTrail and customer-managed KMS keys are part of the baseline in every environment; WAF is added in production only.
AD-15	Roles are fixed at account creation	A tenant administrator creates each staff account with its roles. A staff member signs in to that account and receives exactly the permissions of its roles. There is no role switching in a session; a person who is also a patient uses the patient app with a separate sign-in.

4 Functional requirements

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Functional requirements	Non-functional requirements	Must-haves in Phase 1	Sources traced

4.1 Identity, accounts and access

A person is one human being. A patient profile and any staff memberships hang off the person, and roles are fixed on each account when it is created.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-ACC-01	A patient can create an account with an email address, a password and a mobile number in E.164 format, and must confirm the email address with a one-time code before first sign-in.	Must	PH1	BR-02	Test
FR-ACC-02	A patient can instead sign in with Google. Birthday and phone are requested as optional extra permissions; declining them does not block sign-up. The Google client secret is resolved from Secrets Manager at deploy time.	Should	PH1	BR-02 AD-09	Test
FR-ACC-03	Five roles exist: PATIENT, RECEPTIONIST, CLINICIAN, CLINIC_MANAGER and TENANT_ADMIN. Self-registration exists only for patients; staff and administrator clients have no sign-up path.	Must	PH1	BR-02 BR-07 AD-15	Test
FR-ACC-04	Staff and administrator accounts are created only by a tenant administrator, with their roles, using a temporary password. The fields required for each role are enforced by the service. A password change is forced on first sign-in. Required fields per role are listed in section 7.3.	Must	PH1	BR-07 AD-15	Test
FR-ACC-05	New passwords must be at least 12 characters and contain upper and lower case letters and a number or symbol. A first-sign-in password must differ from the temporary one.	Must	PH1	BR-07	Test
FR-ACC-06	Any user can reset a forgotten password with a code sent to the verified email address.	Must	PH1	BR-02	Test
FR-ACC-07	Every user carries a tenant claim. The authoriser passes tenant, person and roles to handlers, and no handler reads the tenant from a request body or path.	Must	PH1	BR-07 CX-9	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-ACC-08	The authoriser allows a route only when the permission matrix grants it at the needed scope and no guard rule refuses it, verifying the token against cached keys without a runtime identity call. Section 7.3 gives the matrix and the guard rules.	Must	PH1	BR-07 AD-10	Test
FR-ACC-09	A patient can read or cancel only their own appointments. Staff act only within the scope their role grants: own, clinic or tenant.	Must	PH1	BR-01 BR-07	Test
FR-ACC-10	Staff may mark a device as trusted for 12 hours at sign-in. Administrator sessions use a shorter refresh token lifetime than patient sessions.	Could	PH2	BR-07	Inspection
FR-ACC-11	Tenant administrators must use a second factor (TOTP) to sign in.	Should	PH2	CX-13	Test
FR-ACC-12	A user can sign out of all devices, which revokes their refresh tokens.	Should	PH2	CX-13	Test
FR-ACC-13	After sign-up every patient lands on a Complete your profile step that prefills whatever the identity provider returned and requires an explicit confirm. The same step starts empty when nothing was returned.	Must	PH2	AD-09	Test
FR-ACC-14	Date of birth is stored and age is derived when needed. No screen or endpoint accepts an age.	Must	PH2	AD-09	Test
FR-ACC-15	The mobile number is verified by one-time code whatever its source, and the account shows whether it is verified. Only the code sets the verified state.	Must	PH2	AD-09 CX-4	Test
FR-ACC-16	A patient under 18 requires a linked guardian. Consent is recorded with who gave it, when and by what method, and reminders go to the guardian by default. Processing minors' data needs prior authorisation (OI-11).	Must	PH2	AD-09 CX-3 CX-13	Test
FR-ACC-17	A person holds a patient profile and, separately, any staff memberships. A staff member booking for themselves uses the patient app with a patient sign-in; no staff surface books for oneself.	Must	PH2	AD-10	Test
FR-ACC-18	A signed-in account receives exactly the roles set on it at creation. There is no role switching within a session, and a change of roles takes effect at the next sign-in and is audited.	Must	PH1	AD-15 BR-07	Test

4.2 Clinic directory and availability

Patients discover specialists by facts a listing may lawfully carry, and slots are computed on the tenant's grid unit for the chosen type's duration.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-DIR-01	A patient can list the clinicians of their tenant, filtered by specialty and clinic and searchable by name.	Must	PH1	BR-01 BR-06	Test
FR-DIR-02	A patient can list the free start times for one clinician, one date and one appointment type. A start is free when every grid unit of the type's duration and buffer is inside working hours, not locked, not blocked, and at least the type's minimum notice from now.	Must	PH1	BR-01 BR-06 AD-04	Test
FR-DIR-03	Working hours are defined per clinician per weekday on the tenant's grid unit. Phase 1 reads fixed hours from seed configuration.	Must	PH2	CX-10 AD-04	Test
FR-DIR-04	Staff can block and unblock future time. Time that already holds a booking cannot be blocked; the booking must be moved or cancelled first so the patient is notified.	Should	PH2	CX-10	Test
FR-DIR-05	Staff can record a leave period for a clinician. Bookings inside the period go through disruption (FR-STF-10), and the period is excluded from free times.	Should	PH2	CX-10 AD-03	Test
FR-DIR-07	A patient-facing clinician listing shows specialty, qualifications, year of registration, languages and clinic, and nothing else. No score, star, count or rating-derived value appears.	Must	PH2	AD-11	Inspection
FR-DIR-08	The directory is ordered by a seniority band derived from year of registration. The band is never entered by hand.	Should	PH2	AD-11	Test
FR-DIR-09	Where the quality programme is on, a feedback prompt opens after the scheduled end of an attended visit only. It carries structured questions, a free-text review and a safety or conduct flag. The tenant administrator sees all feedback; a clinic manager sees their clinic's aggregate and flagged reviews; a clinician sees their own aggregate and anonymised comments; a receptionist sees none.	Should	PH2	AD-11	Test

4.3 Booking

Three service lines book differently. Specialist bookings lock grid units atomically; general consultations book a place in a session queue; procedures are requested by patients and sized by staff.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-BKG-01	A patient books by first choosing a service line and then an appointment type. For the specialist line they choose a clinician, a date and a free start, with an intake form where the type has one. The API returns 201 with the appointment identifier and a human reference such as APT-4194.	Must	PH1	BR-01 BR-06 AD-01	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-BKG-02	A specialist booking of N grid units writes the appointment and N slot locks in one transaction, each lock conditional on not existing. If any unit is taken the request fails with 409 Conflict and nothing is written.	Must	PH1	BR-06 AD-04	Test
FR-BKG-03	On 409 the client offers the nearest free starts for the same clinician and type.	Should	PH2	BR-06	Demonstration
FR-BKG-04	A booking request carries an Idempotency-Key. A repeated request with the same key within 24 hours returns the original result.	Should	PH1	BR-06	Test
FR-BKG-05	Each type sets its minimum notice and its booking window. The booking and cancel screens show the type's duration and notice period before confirmation.	Should	PH2	CX-5 AD-04	Test
FR-BKG-06	A patient cannot hold two appointments that overlap in time.	Should	PH2	BR-06	Test
FR-BKG-07	Staff can book on behalf of an existing patient, or create a minimal patient record and book in one flow. The booking is attributed to the staff account in the audit trail.	Must	PH2	CX-10 CX-13	Test
FR-BKG-08	The confirmation shows the reference, service line, type, clinic, and the time or arrival window, and offers a calendar file for specialist bookings.	Could	PH2	BR-01	Demonstration
FR-BKG-09	Three service lines exist: general consultation, specialist consultation and procedure. Check-up and follow-up are appointment types inside a line, never lines or top-level booking paths.	Must	PH2	AD-01	Inspection
FR-BKG-10	A clinic manager publishes sessions with a date, a window and the number of rostered GPs. Capacity is derived from rostered GPs and measured throughput and is never typed.	Must	PH2	AD-02	Test
FR-BKG-11	For the general line a patient books a queue ticket in a session. The server assigns the position from the request timestamp. The ticket shows the day and part of day, the facility, an arrival window, the position as Nth of M, and that the patient will be texted if the queue moves. No doctor is named before check-in.	Must	PH2	AD-02 AD-10	Test
FR-BKG-12	Arrival windows are recalculated as the session runs. When a patient's window moves by more than the tenant's threshold they receive an SMS with the new window.	Must	PH2	AD-02 CX-4	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-BKG-13	Walk-ins join the same queue at the desk, so online and physical tickets are one ordered list. A clinician is assigned at check-in, and each session records planned against actual throughput.	Must	PH2	AD-02 CX-10	Test
FR-BKG-14	A patient can request a procedure, which creates a REQUESTED appointment with no time held and no duration asked. Staff size and place it, and it becomes confirmed only when the patient confirms. The patient can follow the request in their appointments.	Should	PH2	AD-05	Test
FR-BKG-15	Each appointment type may carry an intake form. Answers are stored as care context with consent given per booking.	Should	PH2	AD-06	Test

4.4 Viewing and managing visits

Appointments move through a lifecycle that records attendance outcomes, and patients and staff read them at four ranges.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-VIS-01	A patient can list their own appointments and tickets, upcoming and past, ordered by start time.	Must	PH1	BR-01 BR-06	Test
FR-VIS-02	A patient or staff member can open one appointment within their scope.	Must	PH1	BR-01 BR-06	Test
FR-VIS-03	A patient or staff member can cancel a booked appointment. The update is conditional on its state; a second cancel returns 409. Its slot locks are removed so the time is free again.	Must	PH1	BR-01 BR-06	Test
FR-VIS-04	Cancellation deletes the reminder schedule and sends a cancellation email.	Must	PH1	BR-04 BR-09	Test
FR-VIS-05	Staff record a reason for every cancellation and reschedule; the reason is never prefilled and the action cannot complete without it.	Must	PH2	AD-10 CX-7	Test
FR-VIS-06	A cancellation inside the type's notice period is recorded as late.	Should	PH2	CX-7 AD-07	Test
FR-VIS-07	A patient or staff member can reschedule, which releases and locks units atomically and links the new appointment to the old one.	Should	PH2	BR-01	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-VIS-08	The lifecycle includes REQUESTED, DECLINED, BOOKED, CONFIRMED, RESCHEDULED, DISRUPTED, ARRIVED, IN_CONSULTATION, COMPLETED, AWAITING_FEEDBACK, PATIENT_CANCELLED, CLINIC_CANCELLED and NO_SHOW. Closed appointments are retained, never deleted.	Must	PH2	CX-7 AD-03 AD-05 AD-07	Test
FR-VIS-09	Every schedule and history view offers Day, Week, Month and Year. Month is a calendar grid with a density per day; Year is a twelve-month heatmap read as utilisation by staff and as visit history by patients.	Should	PH2	AD-08	Demonstration

4.5 Reminders

A reminder is a one-time schedule created at booking. When it fires, the worker re-reads the appointment and the patient's current contact details.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-REM-01	Booking creates a one-time schedule at the appointment start or arrival window minus the lead time (default 24 hours). The schedule reference is stored on the appointment.	Must	PH1	BR-03 BR-08	Test
FR-REM-02	When the schedule fires, the worker reads the appointment. If it is no longer scheduled it logs the skip and sends nothing.	Must	PH1	BR-08	Test
FR-REM-03	The worker reads the recipient's current phone number at send time (the guardian for a minor, or the proxy where set) and publishes the SMS through SNS as a transactional message.	Must	PH1	BR-03 BR-08 CX-4 CX-3	Test
FR-REM-04	A specialist reminder states the clinic, clinician, local date and time, reference, and how to cancel. A queue ticket reminder states the facility, day, arrival window and position, and names no doctor. No reminder contains the visit reason or any intake answer.	Must	PH1	CX-13 CX-4 AD-02	Inspection
FR-REM-05	Schedule times are computed from clinic local time in the region pack's timezone (Africa/Douala, UTC+1, no daylight saving) and stored in UTC.	Must	PH1	BR-08 AD-13	Test
FR-REM-06	If a booking is made inside the lead time, no reminder is scheduled; the confirmation SMS is sent immediately instead.	Should	PH1	BR-08	Test
FR-REM-07	Lead times are tenant settings. A tenant may enable a second reminder; the default is one reminder at 24 hours and the maximum is two. No trial supports a 2-hour lead time; see OI-06.	Should	PH2	CX-5 CX-6	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-REM-08	Reminder templates exist in French and English for every tenant, and a tenant can add templates in local languages.	Must	PH2	CX-2	Inspection
FR-REM-09	A patient can nominate a proxy recipient who receives reminders in addition to or instead of the patient.	Should	PH2	CX-3	Test
FR-REM-10	A patient can choose a voice call instead of SMS, and an SMS that fails delivery triggers a voice call when the tenant has enabled it. Needs a voice provider; see OI-05.	Should	PH3	CX-1	Test
FR-REM-11	A patient can set notification preferences: channels, language, quiet hours, and opt-in auto-accept of same-day disruption offers.	Should	PH2	CX-2 CX-4 AD-03	Test

4.6 Email and message delivery

Delivery outcome exists only as an event from the provider, so it is captured when it happens.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-MSG-01	A successful booking sends a confirmation email through SES where the patient has an email address.	Must	PH1	BR-04 BR-09	Test
FR-MSG-02	A cancellation sends a cancellation email through SES.	Must	PH1	BR-04 BR-09	Test
FR-MSG-03	Every outbound message is written to a message log with channel, kind, recipient, appointment and state SCHEDULED, SENT, DELIVERED, FAILED or CANCELLED, with a failure reason. Phase 1 records SCHEDULED and SENT; delivery events arrive in Phase 2.	Must	PH1	BR-09 CX-8	Test
FR-MSG-04	An SES configuration set and SNS delivery status logging feed a status handler that updates the message log.	Should	PH2	BR-09 CX-8	Test
FR-MSG-05	Staff see a message log for their clinic in which failed reminders are listed first, so the desk can phone those patients.	Should	PH2	CX-8 CX-10	Demonstration
FR-MSG-06	Tenant administrators see sender identities, delivery rate, bounce and complaint rates, and SMS spend per day.	Should	PH2	CX-8	Demonstration
FR-MSG-07	Emails are sent with an HTML and a plain-text part, in the patient's chosen language.	Should	PH2	CX-2	Inspection

4.7 Staff consoles

Four staff roles see different consoles. The front desk runs the day without clinical context; clinicians see their own patients; managers run a clinic; administrators run the institution.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-STF-01	A clinician can view their own calendar; a receptionist and a clinic manager can view any clinician's day in their clinic.	Must	PH1	BR-06 BR-07	Test
FR-STF-02	The receptionist home shows the clinic's day: appointments across clinicians and the live merged queue, with counts of booked, arrived, waiting and no-show.	Must	PH2	CX-10 AD-02	Demonstration
FR-STF-03	Staff can switch between Day, Week, Month and Year views of load per clinician.	Should	PH2	CX-10 AD-08	Demonstration
FR-STF-04	Staff can record arrival. The action is recorded against the staff account.	Must	PH2	CX-7 CX-13	Test
FR-STF-05	An appointment still scheduled 30 minutes after its end is proposed to staff as a no-show, and becomes NO_SHOW when confirmed.	Must	PH2	CX-7	Test
FR-STF-06	Staff can search the patient register by name, phone or patient number, within their scope.	Should	PH2	CX-10	Test
FR-STF-07	The patient record shows contact details, verification state and visit history within the viewer's scope. Receptionists and clinic managers never see care context.	Should	PH2	CX-13 AD-06	Test
FR-STF-08	Staff can see the clinician roster with shift status and today's load.	Should	PH2	CX-10	Demonstration
FR-STF-09	A staff member sees only data of their own tenant.	Must	PH1	BR-07 CX-9	Test
FR-STF-10	Staff can mark a clinician unavailable for a window. Affected specialist appointments become DISRUPTED and each patient is offered the next matching slots, confirmed with one tap; opted-in patients are moved automatically to same-day slots and told. Queue tickets instead receive moved arrival windows. The action is audited.	Must	PH2	AD-03	Test
FR-STF-11	When the actor is the subject of an appointment, every staff scope is dropped and only the patient permissions apply. Queue positions and slot allocation are assigned by the server; a client-supplied position is rejected.	Must	PH2	AD-10	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-STF-12	A receptionist or clinician cannot move another patient later where they hold an appointment in the same session or adjoining window; the refusal explains why and that a manager can act. A clinic manager may act, and the action is flagged.	Must	PH2	AD-10	Test
FR-STF-13	A clinic manager sees a weekly, read-only list of reschedules and cancellations grouped by actor. Staff priority exists only as a monthly allowance granted by a tenant administrator and shown with its usage.	Should	PH2	AD-10	Demonstration
FR-STF-14	Book for a patient records the channel (walk-in, phone, referral) separately from the service line and type. Walk-in defaults to today's general session; phone requires a callback number; referral captures referring facility, referring clinician and a note, and restricts types to specialist and procedure. Online is set by the system only.	Must	PH2	AD-12 CX-10	Test
FR-STF-15	Staff with procedure permissions see an inbox of procedure requests to size, place or decline.	Should	PH2	AD-05	Demonstration

4.8 Administration console

Tenant administrators are a separate role that owns configuration, accounts and oversight, not day-to-day scheduling, and never reads care context.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-ADM-01	The overview shows accounts, accounts awaiting first sign-in, delivery rate for the last seven days, messages needing attention, capacity gaps for the next week and recent administrator activity.	Should	PH2	CX-8	Demonstration
FR-ADM-02	Administrators can create staff accounts with their roles, resend an invitation, change roles, suspend and re-enable an account, and see which accounts have not yet signed in.	Must	PH2	BR-07 AD-15	Test
FR-ADM-03	Administrators can edit clinic settings: profile, opening hours, grid unit, reminder lead times, languages and message senders.	Must	PH2	CX-2 CX-5 AD-04	Test
FR-ADM-04	Administrators can read the audit log, filter it by actor, action and date, and export it. No audit entry can be edited or deleted.	Must	PH2	CX-13	Test
FR-ADM-05	Administrators see aggregates by default. Opening an individual patient's appointments from the console is itself an audited action.	Should	PH2	CX-13	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-ADM-06	Administrators maintain appointment types (line, duration and buffer in grid units, who may book, notice, cancellation window, fee band, intake form) and intake forms, without a release.	Must	PH2	AD-04 AD-06	Test
FR-ADM-07	Administrators maintain per-practice fee bands taken from the ministerial schedule.	Should	PH2	AD-11 AD-07	Test
FR-ADM-08	Administrators grant and review staff priority allowances with a monthly quota and visible usage.	Could	PH2	AD-10	Demonstration
FR-ADM-09	Administrators see the roles and permissions each role carries, read-only.	Could	PH2	AD-15	Demonstration

4.9 Privacy and data protection

These requirements implement Law No. 2024/017. None is optional for a deployment that processes real patient data.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-PRV-01	At registration the patient is shown who processes their data, for what purposes, which categories, which recipients, for how long, whether it is stored abroad, and their rights, and gives explicit consent. The consent text version and time are stored. Law 2024/017, sections 9 and 21.	Must	PH3	CX-13	Test
FR-PRV-02	A patient under 18 is registered with a guardian whose consent is recorded (FR-ACC-16). The guardian manages the dependant's appointments. Law 2024/017, section 9(3).	Must	PH3	CX-13 AD-09	Test
FR-PRV-03	A patient can download a copy of their personal data in a machine-readable format. Law 2024/017, sections 39 and 43.	Must	PH3	CX-13	Test
FR-PRV-04	A patient can correct their details and request erasure. Erasure removes or anonymises personal data except records the tenant must keep, which are listed to the patient. Law 2024/017, sections 38 and 42.	Must	PH3	CX-13	Test
FR-PRV-05	A patient can withdraw consent, which stops all processing that consent covered. Law 2024/017, sections 9 and 40.	Must	PH3	CX-13	Test
FR-PRV-06	Every read and write of patient personal data by staff or administrators is recorded in the audit trail with actor, action, record, time and source. Law 2024/017, section 27.	Must	PH2	CX-13	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-PRV-07	A breach response procedure exists, with templates for immediate notice to the Data Protection Authority and to affected patients. Law 2024/017, section 22.	Must	PH3	CX-13	Inspection
FR-PRV-08	A register of processing operations and a data protection impact assessment are completed and kept current. Law 2024/017, sections 29 and 33.	Must	PH3	CX-13	Inspection
FR-PRV-09	Retention periods are tenant settings; expired records are deleted or anonymised automatically. Law 2024/017, sections 13 and 28.	Should	PH3	CX-13	Test
FR-PRV-10	Care context is stored apart from the appointment, encrypted under its own key, readable only by the patient and the assigned clinician, never by a receptionist, clinic manager or tenant administrator, and purged on a shorter clock than booking history.	Must	PH2	AD-06 CX-13	Test
FR-PRV-11	Appointment history is retained ten years and shown to the patient for five years by default, both tenant-configurable. Screens that state retention distinguish retained from shown and state care context separately. Unconfirmed; see OI-12.	Should	PH2	AD-08 CX-13	Inspection

4.10 Offline and low connectivity

Coverage gaps are expected. The clients degrade rather than stop.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-OFF-01	The patient app shows upcoming appointments and tickets offline from its last sync, with a banner saying when data was last updated.	Should	PH2	CX-11	Demonstration
FR-OFF-02	Bookings and cancellations require a connection; the client says so instead of queuing them.	Must	PH2	CX-11	Test
FR-OFF-03	The staff app queues arrival and no-show marks made offline and sends them when the connection returns. A queued mark on an appointment cancelled meanwhile is rejected and shown to the staff member.	Should	PH3	CX-11	Test

4.11 Measurement and the fee ledger

The first clinic's non-attendance baseline is a deliverable of the system. Every decision a refund depends on is recorded from Phase 2, and no money moves until a payment provider is added.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-MSR-01	The system reports, per clinic and per week, appointments booked, attended, no-show, cancelled and late-cancelled, from the first day of use.	Must	PH2	CX-7	Test
FR-MSR-02	Attendance can be broken down by channel and by whether the reminder was delivered, failed or not sent.	Should	PH2	CX-7 CX-8 AD-12	Test
FR-MSR-03	Reports can be exported as CSV without patient identifiers.	Could	PH3	CX-7 CX-13	Test
FR-MSR-04	No screen or report states a currency saving attributed to reminders.	Must	PH2	CX-12	Inspection
FR-MSR-05	Every appointment writes a fee ledger entry at closure holding the fee band, the amount in XAF, the outcome and the refund entitlement.	Must	PH2	AD-07	Test
FR-MSR-06	Outcomes map to entitlement as follows: attended NONE; cancelled in window FULL; cancelled late PARTIAL; no-show NONE; clinic cancelled FULL; clinic rescheduled NONE. The patient cancel flow states the entitlement band before the confirm control.	Must	PH2	AD-07	Test
FR-MSR-07	No payment, card entry, wallet or settlement function exists before Phase 3; entitlement is a record, never money moved or owed.	Must	PH2	AD-07	Inspection

4.12 Tenancy and configuration

Tenant-prefixed keys and the tenant claim are paid for now because they cannot be retrofitted cheaply. Country specifics come from a region pack.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-TEN-01	Every storage key begins with the tenant identifier, except country-level region packs.	Must	PH1	CX-9	Inspection
FR-TEN-02	A tenant sets a grid unit (default 10 minutes), and every appointment type carries its own duration and buffer in grid units. No single tenant-wide slot length exists.	Must	PH2	CX-9 AD-04	Test
FR-TEN-03	Tenant onboarding commercial terms, per-tenant branding and billing are out of scope.	Must	PH1	CX-9	Inspection
FR-TEN-04	No-show strikes are off by default. When a tenant sets a threshold, a patient over it within a rolling 180 days can book only within 24 hours, is told why, and is never locked out.	Should	PH2	AD-07	Test
FR-TEN-05	A tenant can switch the quality programme off. The ratingAffectsFee flag exists, defaults to off, and is refused by the service wherever the region pack forbids it.	Should	PH2	AD-11	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-TEN-06	A tenant chooses a region pack at onboarding. The pack supplies currency, phone rules, languages, timezone, public holidays, address and national ID formats, SMS sender policy, emergency number and default fees. The Cameroon pack uses XAF, +237, French and English, Africa/Douala, the ten regions and CNI.	Must	PH2	AD-13	Test
FR-TEN-07	The region pack is locked once the tenant has its first appointment, and the setting shows why it is locked.	Should	PH2	AD-13	Test
FR-TEN-08	Every seed, demo and sample value is generated from the active region pack, and no real personal data exists outside an authorised production environment.	Must	PH1	AD-13 CX-14	Inspection

4.13 Interoperability

Atria does not integrate with the national platform in this release, but its data must not make integration a rebuild.

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
FR-INT-01	Clinics, clinicians, patients, availability, sessions, slots and appointments map onto FHIR R4 Location, Practitioner, Patient, Schedule, Slot and Appointment, and the mapping is documented.	Should	PH3	CX-15	Inspection
FR-INT-02	Staff can export one appointment as a FHIR R4 Appointment resource in JSON.	Could	PH3	CX-15	Test

Retired identifiers are not reused.

FR-DIR-06 (ratings hidden until a source exists) is replaced by FR-DIR-07 to FR-DIR-09.

5 External interfaces

5.1 User interfaces

Six surfaces carry the product: patient phone, patient tablet and patient desktop; staff phone and staff desktop, shared by the four staff roles with role-specific navigation; and administration on desktop web. Each account signs in to its own role and sees only that role's navigation.

5.2 Application programming interface

REST over HTTPS, JSON bodies, stage `v1`. Every request carries a Cognito token as `Authorization: Bearer`. Errors return `{"error":{"code","message","rule","correlationId"}}` with 400, 401, 403, 404, 409, 422, 429 or 500.

IDENTITY AND ACCOUNTS

METHOD	PATH	PERMISSION	PURPOSE
POST	/me/profile	patient.update (own)	Complete your profile: confirm names and date of birth, set language
POST	/me/phone/code · /me/phone/verify	patient.update (own)	Send and check the one-time code that sets phoneVerifiedAt
POST	/me/guardian	patient.update (own)	Link a guardian for a patient under 18 and record consent method
GET · PUT	/me/preferences	patient.update (own)	Channels, lead times, proxy recipient, auto-accept for same-day offers
GET · POST	/me/consents · /me/export · /me/erasure	patient.read (own)	Data subject rights
POST	/admin/staff	staff.create	Create a staff account with its roles; fields per role are enforced in the service
PATCH	/admin/staff/{id}/roles	staff.update_roles	Change the roles on an existing account; takes effect at next sign-in
POST	/admin/staff/{id}/suspend	staff.suspend	Suspend an account and revoke its refresh tokens
PUT	/admin/staff/{id}/priority-allowance	priority_allowance.grant	Grant or change a monthly priority quota

DIRECTORY AND CAPACITY

METHOD	PATH	PERMISSION	PURPOSE
GET	/clinicians?specialty=&clinicId=	any signed-in	Directory ordered by seniority band. Specialty, qualifications, registration year, languages, clinic; never a score

METHOD	PATH	PERMISSION	PURPOSE
GET	/clinicians/{id}/slots?date=&typeId=	any signed-in	Free start units for a type's duration and buffer
GET · PUT	/clinicians/{id}/availability	availability.read · availability.write	Weekly template on the grid unit
POST	/clinicians/{id}/unavailability	session.disrupt	Mark a clinician unavailable for a window; starts disruption
GET · POST	/sessions	availability.read · session.publish	List or publish sessions with window and rostered GPs; capacity is derived
POST	/sessions/{id}/disrupt	session.disrupt	Shift every ticket's arrival window and notify

BOOKING AND THE DAY

METHOD	PATH	PERMISSION	PURPOSE
POST	/appointments	appointment.create	Specialist booking: type, clinician, start, channel, intake answers and consent. Locks N units atomically
POST	/sessions/{id}/tickets	appointment.create	General line: issue a queue ticket; position assigned by the server
GET	/tickets/{id}	appointment.read	Live position and arrival window
POST	/procedure-requests	procedure.request	Raise a REQUESTED appointment with no time held
POST	/procedure-requests/{id}/placement	procedure.size_and_place	Size and place a request for the patient to confirm
POST	/appointments/{id}/confirm	appointment.read (own)	Patient confirms a placed procedure or a disruption offer
GET	/patients/me/appointments?range=&from=	appointment.read (own)	Day, week, month or year; five-year default horizon
GET	/appointments/{id}	appointment.read	One appointment within scope
PATCH	/appointments/{id}	appointment.reschedule	Reschedule; staff must give a reason
DELETE	/appointments/{id}	appointment.cancel	Cancel; returns the entitlement band; staff must give a reason
POST	/appointments/{id}/arrival · /no-show	appointment.check_in · appointment.mark_no_show	Record attendance outcomes
POST	/sessions/{id}/next	queue.assign_next	Assign the next ticket to the calling clinician

METHOD	PATH	PERMISSION	PURPOSE
GET · PUT	/appointments/{id}/care-context	care_context.read · care_context.write	Care context for the patient and the assigned clinician only
POST	/appointments/{id}/referral	appointment.create	Referring facility and clinician, with a presigned upload for the note

PEOPLE, QUALITY AND MONEY

METHOD	PATH	PERMISSION	PURPOSE
GET · POST	/patients	patient.read · patient.create	Search the register; quick-create a patient at the desk
GET · PATCH	/patients/{id}	patient.read · patient.update	Patient record within scope; strike count when strikes are on
POST	/appointments/{id}/feedback	feedback.submit	Structured answers, free text and a safety or conduct flag; attended visits only
GET	/feedback?clinicId=&clinicianId=&from=	feedback.read	Scoped: own aggregate, clinic aggregate and flagged reviews, or tenant-wide
GET	/fee-ledger?clinicId=&from=	fee_ledger.read	Fee band, XAF amount, outcome and entitlement per appointment

INSTITUTION AND OVERSIGHT

METHOD	PATH	PERMISSION	PURPOSE
GET · POST · PATCH	/admin/clinics · /admin/appointment-types · /admin/intake-forms	clinic.manage · appointment_type.manage	Institution configuration
GET · POST · PATCH	/admin/fee-bands	fee_band.manage	Per-practice baselines from the ministerial schedule
PUT	/admin/region-pack	region_pack.choose	Choose the region pack; refused once the first appointment exists
PUT	/admin/feature-flags	feature_flag.set	Quality programme, strikes; ratingAffectsFee refused where the pack forbids it
GET	/audit?actor=&action=&from=&to=	audit.read	Read-only audit trail
GET	/reports/attendance?by=channel · /reports/reschedules/weekly	report.read	Attendance and no-show by channel; weekly reschedules grouped by actor

METHOD	PATH	PERMISSION	PURPOSE
GET · POST	/messages?status=	message.send	Message log with failures first; resend

5.3 Messaging interfaces

INTERFACE	REQUIREMENT
SMS	SNS, transactional type, E.164 recipients, sender ID registered per network. Templates in French and English fit one GSM-7 segment of 160 characters where possible. A provider adapter allows a third-party SMS or voice provider whose key is held in Secrets Manager (BR-05).
Email	SES from a verified domain with DKIM, SPF and DMARC, through a configuration set that publishes delivery, bounce and complaint events.
Delivery events	SES events and SNS delivery status logs arrive on one topic consumed by the message status handler.

5.4 Identity interfaces

- Cognito user pool with separate patient, staff and administrator app clients. Only the patient client allows sign-up.
- Google as an identity provider on the patient client only, authorisation code flow, scopes `openid email profile`, with birthday and phone as optional extra permissions.
- Tokens carry the tenant, the person and the account's roles, added at token generation.

6 Non-functional requirements

6.1 Performance

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-PERF-01	95 per cent of read requests complete in under 800 ms and 95 per cent of booking requests in under 1,500 ms, measured at the API over a week of normal load.	Should	PH1	BR-06	Test
NFR-PERF-02	A reminder is published within 2 minutes after its scheduled time, and never before it.	Must	PH1	BR-08	Test
NFR-PERF-03	The patient web app's first route loads under 200 KB of compressed JavaScript and is usable within 5 seconds on a 1.6 Mbps connection.	Should	PH2	CX-4	Test
NFR-PERF-04	Every list endpoint is paginated at 50 items or fewer.	Must	PH1	BR-06	Inspection

6.2 Availability and reliability

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-REL-01	The booking API is available 99.5 per cent of each calendar month, excluding announced maintenance.	Should	PH2	BR-06	Analysis
NFR-REL-02	Point-in-time recovery is enabled on every table (recovery point within 5 minutes); a full restore completes within 4 hours.	Must	PH2	CX-13	Test
NFR-REL-03	Handlers are idempotent. Asynchronous handlers retry with backoff.	Must	PH1	BR-08 BR-09	Test
NFR-REL-04	Every asynchronous consumer reads from SQS with a dead letter queue that retains messages 14 days, and redrive is a documented, tested procedure.	Must	PH2	AD-14 BR-08	Test

6.3 Security

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-SEC-01	All traffic uses TLS 1.2 or later. The web app sends HSTS.	Must	PH1	CX-13	Test
NFR-SEC-02	Data at rest is encrypted with customer-managed KMS keys, with care context on its own key.	Must	PH1	CX-13 AD-06	Inspection
NFR-SEC-03	Every secret, including third-party SMS or voice keys and the Google client secret, lives in Secrets Manager. No secret appears in code, the repository, a function environment variable or a template. CI scans every push for secrets.	Must	PH1	BR-05 BR-10	Inspection

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-SEC-04	Each function has its own IAM role granting only the actions and resources it uses.	Must	PH1	BR-07	Inspection
NFR-SEC-05	Request bodies are validated against a schema at the API and again in the handler. The API is rate limited per client and per caller.	Must	PH1	BR-06	Test
NFR-SEC-06	Logs never contain full phone numbers, email addresses or names; they are masked.	Must	PH1	CX-13	Inspection
NFR-SEC-07	The web and mobile clients meet OWASP ASVS level 2 for authentication, session management and access control.	Should	PH3	CX-13	Analysis
NFR-SEC-08	CloudTrail records every control-plane action in every environment. In production the web app and the API sit behind AWS WAF; development and test environments, which hold synthetic data only, run without it to limit cost.	Must	PH2	AD-14 CX-13	Inspection

6.4 Compliance

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-CMP-01	No real patient data is processed until the Data Protection Authority's authorisation is held.	Must	PH3	CX-13 CX-14	Inspection
NFR-CMP-02	The production region is documented and the authorisation to transfer data abroad is held before go-live.	Must	PH3	CX-14	Inspection
NFR-CMP-03	The controller's annual security report to the Authority can be produced from system records.	Should	PH3	CX-13	Inspection
NFR-CMP-04	Development runs in us-east-1 on synthetic data only; the region is a per-environment parameter.	Must	PH1	CX-14 AD-13	Inspection

6.5 Usability and accessibility

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-UX-01	Web surfaces meet WCAG 2.2 level AA. Flutter screens expose semantic labels for every control.	Must	PH2	CX-4	Test
NFR-UX-02	Touch targets on phone layouts are at least 46 by 46 px; interfaces remain usable with text scaled to 200 per cent; reduced-motion settings turn transitions off.	Must	PH2	CX-4	Test
NFR-UX-03	A patient who has chosen a clinician and type can complete a booking in at most four actions, intake excluded.	Should	PH2	BR-01	Test

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-UX-04	Every error message says what went wrong and what to do next.	Must	PH1	BR-06	Inspection

6.6 Localisation

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-L10N-01	Every patient and staff surface is available in French and English, selectable at any time.	Must	PH2	CX-2	Inspection
NFR-L10N-02	Money, phone numbers, dates, times, addresses and ID formats follow the region pack: for Cameroon XAF without decimals, +237, Africa/Douala, and the 24-hour clock for staff.	Must	PH1	CX-4 AD-13	Inspection

6.7 Maintainability and delivery

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-MNT-01	Every AWS resource is defined in AWS CDK v2 in Python, one stack per concern: network, data, identity, API, async, observability and cost guard. Nothing is created by hand in the console.	Must	PH1	BR-11 AD-14	Inspection
NFR-MNT-02	The code lives in one repository with infrastructure, backend, web, mobile and seed packages. Each feature is developed on its own branch and merged by reviewed pull request.	Must	PH1	BR-11	Inspection
NFR-MNT-03	Every pull request runs lint, type checks, unit tests, a CDK synth with security rule checks, Flutter analyze and test, and the web build and test. A failing check blocks the merge.	Must	PH1	BR-11	Inspection
NFR-MNT-04	Merges to the integration branch deploy to development automatically. Staging and production deploy from a tagged release after manual approval.	Must	PH1	BR-11	Demonstration
NFR-MNT-05	Domain logic has at least 80 per cent line coverage, including every guard rule and entitlement outcome. Each endpoint has an integration test, and booking, queueing and cancellation have end-to-end tests on both clients.	Must	PH1	BR-06	Test

6.8 Observability and cost

ID	REQUIREMENT	PRIORITY	PHASE	TRACE	VERIFY
NFR-OBS-01	Handlers write structured JSON logs with a correlation identifier that follows a request across functions, and the booking path is traced with X-Ray.	Must	PH1	BR-06	Inspection
NFR-OBS-02	Alarms fire on API 5xx rate, reminder failures, dead letter queue depth, SES bounce rate and daily SMS spend.	Must	PH2	BR-08 BR-09 CX-8	Test
NFR-OBS-03	SMS spend is capped per tenant per day.	Should	PH2	CX-8	Test
NFR-OBS-04	The outbox publishes to a FIFO queue keyed on appointment identifier, and EventBridge delivers into SQS rather than directly into functions.	Must	PH2	AD-14	Inspection
NFR-OBS-05	Log groups have explicit retention, alarms also cover function and table throttles, and each environment has one dashboard.	Should	PH2	AD-14	Inspection
NFR-OBS-06	AWS Budgets alert at 50, 80 and 100 per cent of each environment's budget, and Cost Anomaly Detection is enabled.	Should	PH1	AD-14	Test

7 Data and access requirements

7.1 Entities

The system stores 26 entities in five groups. Care context is stored apart from the appointment, on its own key.

ENTITY	GROUP	PURPOSE	ATTRIBUTES
REGION_PACK	Tenancy and configuration	The locale a tenant operates in. Supplies everything country specific so a second country is a data change, not a code change.	12
TENANT	Tenancy and configuration	An institution using Atria. Every record below belongs to exactly one.	9
CLINIC	Tenancy and configuration	A physical site within a tenant where appointments take place.	6
APPOINTMENT_TYPE	Tenancy and configuration	What is being booked. Carries its own duration, which is the change that removes the single tenant wide slot length.	15
INTAKE_FORM	Tenancy and configuration	The questions asked at booking, chosen by appointment type.	5
FEE_BAND	Tenancy and configuration	The baseline fee for a practice, not for an individual. Set by the tenant administrator from the ministerial schedule.	6
PERSON	Identity and consent	One human being. Holds no role. This split is what lets a staff member be a patient without being able to act on their own appointment as staff.	9
PATIENT_PROFILE	Identity and consent	A person as a patient of one tenant.	8
STAFF_MEMBERSHIP	Identity and consent	A person's employment at one clinic, carrying their roles.	7
CLINICIAN_PROFILE	Identity and consent	The bookable part of a staff membership. Only the facts Article 7 permits in a listing appear on the patient side.	8
CONSENT_RECORD	Identity and consent	Who consented to what, when and how. Required by Law 2024/017 and the evidence a pilot has to produce.	9
PRIORITY_ALLOWANCE	Identity and consent	An explicit, bounded, visible staff booking privilege. The alternative to a loophole.	5
AVAILABILITY_TEMPLATE	Scheduling	A clinician's recurring working hours.	8
AVAILABILITY_EXCEPTION	Scheduling	A one off change. Marking a clinician unavailable here is what starts a disruption.	7
SESSION	Scheduling	A published block of general consultation capacity. Patients book a place in it rather than a named doctor.	12

ENTITY	GROUP	PURPOSE	ATTRIBUTES
QUEUE_TICKET	Scheduling	A patient's place in a session. Position is assigned by the server from the request timestamp, which is	9
SLOT_LOCK	Scheduling	The item whose existence prevents double booking. One per grid unit, all written in a single transaction so an N unit appointment is still atomic.	4
APPOINTMENT	Scheduling	The central record. Holds either a clinician and a time, or a session and a ticket, never both.	16
APPOINTMENT_EVENT	Scheduling	Every transition, with who caused it and why. Drives the patient timeline and satisfies the audit obligation.	9
REFERRAL	Scheduling	Where a specialist appointment came from. The normal door into the specialist line, matching the health pyramid.	7
CARE_CONTEXT	Scheduling	What the patient chose to send ahead of one visit. Deliberately its own entity, its own key and its own shorter retention.	8
FEE_LEDGER_ENTRY	Accountability and quality	What was owed and what the outcome entitles the patient to. Nothing settles until a payment provider arrives in Phase 3.	9
FEEDBACK	Accountability and quality	Collected after the scheduled end, only where attendance was recorded. Seen by the tenant administrator. Never shown on a patient facing directory.	11
REMINDER	Messaging and audit	One scheduled send. Queued through SQS so a failure is parked and redrivable rather than lost.	8
MESSAGE_LOG	Messaging and audit	What was actually sent, to whom, and what the provider said.	8
AUDIT_ENTRY	Messaging and audit	Every privileged action. A legal obligation here, not a convenience.	10

7.2 Classification and retention

DATA	CLASSIFICATION	DEFAULT RETENTION
Appointment history and ledger	Sensitive. Service line and specialty can reveal a health condition.	Ten years retained, five years shown to the patient (OI-12)
Care context	Sensitive, special key	Purged after the visit's review window, default 90 days
Contact details	Personal	Life of the account, then erased on request
Message log	Personal	12 months
Audit trail	Personal, integrity-protected	Ten years

DATA	CLASSIFICATION	DEFAULT RETENTION
Consent records	Personal, evidential	Life of the account plus five years
Slot locks	Operational, no personal data	Removed with the slot

Retention is proposed, not decided.

No statutory retention period for Cameroonian medical records was found. Law No. 2024/017 places retention in the Authority's authorisation. FR-PRV-09 and FR-PRV-11 make the values configurable so the authorised values can be applied without a release.

7.3 Access control

Permissions by role and scope. *own* is the actor's own records, *clinic* every record in their clinic, *tenant* every record in the tenant.

PERMISSION	PATIENT	RECEP.	CLINICIAN	MANAGER	ADMIN
appointment.read	own	clinic	own	clinic	tenant
appointment.create	own	clinic	clinic	clinic	none
appointment.reschedule	own	clinic	own	clinic	none
appointment.cancel	own	clinic	own	clinic	none
appointment.check_in	none	clinic	own	clinic	none
appointment.mark_no_show	none	clinic	own	clinic	none
procedure.size_and_place	none	none	own	clinic	none
procedure.request	own	clinic	clinic	clinic	none
care_context.read	own	none	own	none	none
care_context.write	own	none	own	none	none
availability.read	none	clinic	clinic	clinic	tenant
availability.write	none	none	own	clinic	none
session.publish	none	none	none	clinic	none
session.disrupt	none	clinic	clinic	clinic	none
queue.assign_next	none	clinic	clinic	clinic	none
patient.read	own	clinic	own	clinic	tenant

PERMISSION	PATIENT	RECEP.	CLINICIAN	MANAGER	ADMIN
patient.create	own	clinic	clinic	clinic	none
patient.update	own	clinic	none	clinic	none
staff.create	none	none	none	none	tenant
staff.update_roles	none	none	none	none	tenant
staff.suspend	none	none	none	none	tenant
priority_allowance.grant	none	none	none	none	tenant
clinic.manage	none	none	none	none	tenant
appointment_type.manage	none	none	none	none	tenant
fee_band.manage	none	none	none	none	tenant
region_pack.choose	none	none	none	none	tenant
feature_flag.set	none	none	none	none	tenant
feedback.submit	own	none	none	none	none
feedback.read	own	none	own	clinic	tenant
fee_ledger.read	own	clinic	none	clinic	tenant
audit.read	none	none	none	clinic	tenant
report.read	none	none	own	clinic	tenant
message.send	none	clinic	own	clinic	tenant

GUARD RULE	EFFECT
SELF_SUBJECT_DROPS_STAFF_SCOPE	Where the actor's personId equals the subject's personId, every staff scope is dropped and only the PATIENT row applies. A staff member can never act on their own appointment through staff tooling.
SERVER_ASSIGNED_ORDER	Queue position and slot allocation are derived from the request timestamp on the server. A client supplied position or ordering field is rejected, not ignored.
NO_SELF_BENEFITING_RESCHEDULE	A staff action that moves another patient later is refused where the actor holds an appointment in the same session or the adjoining slot window. A CLINIC_MANAGER may still perform it, which keeps genuine cases possible.

GUARD RULE	EFFECT
REASONED_AND_AUDITED	Every staff initiated reschedule or cancellation requires a reason and writes an AUDIT_ENTRY. Where the actor holds an appointment in the affected window the entry carries POTENTIAL_SELF_BENEFIT for the clinic manager's weekly list.
BOUNDED_PRIORITY	Staff priority exists only as a PRIORITY_ALLOWANCE granted by a tenant administrator, with a monthly quota, visible on the admin dashboard. There is no implicit staff priority anywhere.
REGION_REFUSES_FEE_LINK	Setting tenant.ratingAffectsFee is refused where the region pack's allowsRatingFeeLink is false. Enforced in the region pack, not the screen, so it cannot be bypassed through the API.
TENANT_ADMIN_NEVER_READS_CARE_CONTEXT	care_context.read is none for TENANT_ADMIN and there is no override. The administrative role and the clinical role are separated on purpose.

ROLE	REQUIRED AT ACCOUNT CREATION	HOME AFTER SIGN-IN
PATIENT	registration details	Upcoming visits
RECEPTIONIST	givenName, familyName, phoneE164, clinicId	Today at this clinic
CLINICIAN	givenName, familyName, phoneE164, clinicId, specialty, registrationYear, ordreNumber, languages	My calendar
CLINIC_MANAGER	givenName, familyName, phoneE164, clinicId	Clinic overview and rota
TENANT_ADMIN	givenName, familyName, phoneE164, email	Tenant overview

8 Acceptance criteria

A phase is accepted when every test below for that phase passes against a deployed environment.

BR-06 **Book, retrieve and cancel end to end**

- **GIVEN** a signed-in patient and a free specialist start for a 30-minute type on a 10-minute grid
- **WHEN** they book it
- **THEN** the API returns 201, the appointment appears in their list, three slot locks exist, and a booking by anyone of any overlapping start returns 409
- **AND** cancelling returns 200 with the entitlement band, the time is free again, and a second cancel returns 409

BR-07 **Roles separate access**

- **GIVEN** tokens for a patient and for each staff role
- **WHEN** each calls every route
- **THEN** each succeeds only where the matrix and guard rules allow and receives 403 elsewhere
- **AND** no staff account can be created through the patient client, and a staff member cannot act on their own appointment through staff routes

BR-08 **The reminder fires ahead of the appointment**

- **GIVEN** an appointment booked 26 hours ahead with a 24-hour lead time
- **WHEN** the schedule fires
- **THEN** an SMS is published to the current number within 2 minutes of the scheduled time and before the appointment
- **AND** for an appointment cancelled before that time, nothing is published and the skip is logged

BR-09 **Email is sent reliably**

- **GIVEN** 20 bookings and 20 cancellations to the SES mailbox simulator
- **WHEN** they complete
- **THEN** 40 emails are accepted and 40 message log entries show SENT

BR-10 **Secrets Manager holds every key**

- **GIVEN** the synthesised templates, function configurations and repository
- **WHEN** they are scanned
- **THEN** no secret value is found, and every provider key resolves from Secrets Manager

BR-11 **Deployed through infrastructure as code**

GIVEN an empty AWS account

WHEN the pipeline deploys every stack

THEN the tests above pass with no manual console step

AD-02 **Queue order is fair and live**

GIVEN a session with 24 places and 10 tickets

WHEN two patients and a walk-in request tickets at the same moment

THEN positions follow the server's request timestamps, the walk-in appears in the same list, and a client-supplied position is rejected

AND when the session runs 40 minutes late every later ticket's window moves and those patients receive an SMS

AD-06 **Care context stays with the patient and the clinician**

GIVEN a booking with intake answers

WHEN the receptionist, clinic manager, tenant administrator, assigned clinician and patient request its care context

THEN only the assigned clinician and the patient receive it; the others receive 403

AD-11 **Ratings never touch fees in Cameroon**

GIVEN a tenant on the Cameroon region pack

WHEN an administrator sets ratingAffectsFee to true through the API

THEN the request is refused with 422 and the stored value stays false

AND no directory response contains a score for any clinician

9 Traceability

Computed from the trace field of every requirement. Every objective, context requirement and design decision is covered by at least one requirement.

SOURCE	STATEMENT	COVERED BY
BR-01	Provide a REST API through which patients book, view and cancel appointments.	FR-ACC-09 FR-DIR-01 FR-DIR-02 FR-BKG-01 FR-BKG-08 FR-VIS-01 FR-VIS-02 FR-VIS-03 FR-VIS-07 NFR-UX-03
BR-02	Authenticate patients and staff separately using Cognito.	FR-ACC-01 FR-ACC-02 FR-ACC-03 FR-ACC-06
BR-03	Send appointment reminders through Amazon SNS.	FR-REM-01 FR-REM-03
BR-04	Send confirmation and cancellation emails through Amazon SES.	FR-VIS-04 FR-MSG-01 FR-MSG-02
BR-05	Store third-party service keys, such as an SMS or verification API key, in AWS Secrets Manager.	NFR-SEC-03
BR-06	Booking, retrieval and cancellation endpoints work end to end.	FR-DIR-01 FR-DIR-02 FR-BKG-01 FR-BKG-02 FR-BKG-03 FR-BKG-04 FR-BKG-06 FR-VIS-01 FR-VIS-02 FR-VIS-03 FR-STF-01 NFR-PERF-01 NFR-PERF-04 NFR-REL-01 NFR-SEC-05 NFR-UX-04 NFR-MNT-05 NFR-OBS-01
BR-07	Cognito distinguishes patient and staff access correctly.	FR-ACC-03 FR-ACC-04 FR-ACC-05 FR-ACC-07 FR-ACC-08 FR-ACC-09 FR-ACC-10 FR-ACC-18 FR-STF-01 FR-STF-09 FR-ADM-02 NFR-SEC-04
BR-08	The SNS reminder fires ahead of the scheduled appointment time.	FR-REM-01 FR-REM-02 FR-REM-03 FR-REM-05 FR-REM-06 NFR-PERF-02 NFR-REL-03 NFR-REL-04 NFR-OBS-02
BR-09	SES sends confirmation and cancellation emails reliably.	FR-VIS-04 FR-MSG-01 FR-MSG-02 FR-MSG-03 FR-MSG-04 NFR-REL-03 NFR-OBS-02
BR-10	Secrets Manager holds every sensitive key; nothing is hardcoded in environment configuration.	NFR-SEC-03
BR-11	The system is deployed through infrastructure as code, with a report and an architecture diagram.	NFR-MNT-01 NFR-MNT-02 NFR-MNT-03 NFR-MNT-04

SOURCE	STATEMENT	COVERED BY
CX-1	Voice call fallback for reminders, selectable per patient	FR-REM-10
CX-2	Local-language message templates, configurable per tenant	FR-REM-08 FR-REM-11 FR-MSG-07 FR-ADM-03 NFR-L10N-01

SOURCE	STATEMENT	COVERED BY
CX-3	A nominated proxy recipient on the patient record	FR-ACC-16 FR-REM-03 FR-REM-09
CX-4	SMS-first delivery, with the web and app channels optional	FR-ACC-15 FR-BKG-12 FR-REM-03 FR-REM-04 FR-REM-11 NFR-PERF-03 NFR-UX-01 NFR-UX-02 NFR-L10N-02
CX-5	Reminder lead times configurable per tenant	FR-BKG-05 FR-REM-07 FR-ADM-03
CX-6	Reminder count capped at two by default	FR-REM-07
CX-7	Per-clinic baseline non-attendance measured from first deployment	FR-VIS-05 FR-VIS-06 FR-VIS-08 FR-STF-04 FR-STF-05 FR-MSR-01 FR-MSR-02 FR-MSR-03
CX-8	Delivery status recorded per message	FR-MSG-03 FR-MSG-04 FR-MSG-05 FR-MSG-06 FR-ADM-01 FR-MSR-02 NFR-OBS-02 NFR-OBS-03
CX-9	Tenant boundary modelled on the institution or health district	FR-ACC-07 FR-STF-09 FR-TEN-01 FR-TEN-02 FR-TEN-03
CX-10	Staff-side scheduling usable without patient-side adoption	FR-DIR-03 FR-DIR-04 FR-DIR-05 FR-BKG-07 FR-BKG-13 FR-MSG-05 FR-STF-02 FR-STF-03 FR-STF-06 FR-STF-08 FR-STF-14
CX-11	Offline-tolerant staff client with queued writes	FR-OFF-01 FR-OFF-02 FR-OFF-03
CX-12	No currency-denominated savings claim in product material	FR-MSR-04
CX-13	Compliance with Law No. 2024/017 on personal data protection	FR-ACC-11 FR-ACC-12 FR-ACC-16 FR-BKG-07 FR-REM-04 FR-STF-04 FR-STF-07 FR-ADM-04 FR-ADM-05 FR-PRV-01 FR-PRV-02 FR-PRV-03 FR-PRV-04 FR-PRV-05 FR-PRV-06 FR-PRV-07 FR-PRV-08 FR-PRV-09 FR-PRV-10 FR-PRV-11 FR-MSR-03 NFR-REL-02 NFR-SEC-01 NFR-SEC-02 NFR-SEC-06 NFR-SEC-07 NFR-SEC-08 NFR-CMP-01 NFR-CMP-03
CX-14	Data Protection Authority authorisation, including for hosting abroad	FR-TEN-08 NFR-CMP-01 NFR-CMP-02 NFR-CMP-04
CX-15	Data modelled for exchange as HL7 FHIR resources	FR-INT-01 FR-INT-02

SOURCE	DECISION	COVERED BY
AD-01	Three service lines	FR-BKG-01 FR-BKG-09

SOURCE	DECISION	COVERED BY
AD-02	Book a place in a session	FR-BKG-10 FR-BKG-11 FR-BKG-12 FR-BKG-13 FR-REM-04 FR-STF-02
AD-03	Disruption offers, it does not impose	FR-DIR-05 FR-VIS-08 FR-REM-11 FR-STF-10
AD-04	Grid unit plus typed duration	FR-DIR-02 FR-DIR-03 FR-BKG-02 FR-BKG-05 FR-ADM-03 FR-ADM-06 FR-TEN-02
AD-05	Procedures are staff-sized	FR-BKG-14 FR-VIS-08 FR-STF-15
AD-06	Intake per appointment type	FR-BKG-15 FR-STF-07 FR-ADM-06 FR-PRV-10 NFR-SEC-02
AD-07	Fee ledger without moving money	FR-VIS-06 FR-VIS-08 FR-ADM-07 FR-MSR-05 FR-MSR-06 FR-MSR-07 FR-TEN-04
AD-08	Four range views and a history horizon	FR-VIS-09 FR-STF-03 FR-PRV-11
AD-09	Date of birth and a verified phone	FR-ACC-02 FR-ACC-13 FR-ACC-14 FR-ACC-15 FR-ACC-16 FR-PRV-02
AD-10	One person, separate profiles, fair queues	FR-ACC-08 FR-ACC-17 FR-BKG-11 FR-VIS-05 FR-STF-11 FR-STF-12 FR-STF-13 FR-ADM-08
AD-11	Ratings stay off prices and off names	FR-DIR-07 FR-DIR-08 FR-DIR-09 FR-ADM-07 FR-TEN-05
AD-12	Channel and type are two axes	FR-STF-14 FR-MSR-02
AD-13	Region pack	FR-REM-05 FR-TEN-06 FR-TEN-07 FR-TEN-08 NFR-CMP-04 NFR-L10N-02
AD-14	Queues, dead letters, alarms and a budget	NFR-REL-04 NFR-SEC-08 NFR-MNT-01 NFR-OBS-04 NFR-OBS-05 NFR-OBS-06
AD-15	Roles are fixed at account creation	FR-ACC-03 FR-ACC-04 FR-ACC-18 FR-ADM-02 FR-ADM-09

10 Open issues and risks

Items that are known but not yet resolved. Each has an owner and a point by which it must close.

ID	ISSUE	IMPACT	RESOLUTION	OWNER	BY
OI-02	Production region and the authorisation to transfer data abroad.	Go-live blocked without authorisation.	Choose the production region before Phase 3; apply with the processing request.	Team, clinic	Before PH3
OI-03	The Authority's authorisation procedure is left to regulation and was not located.	Unknown lead time for a pilot.	Ask the Ministry and the clinic's counsel; track publication.	Clinic	Before PH3
OI-04	SMS sender ID registration for MTN, Orange and Camtel differs by region and is untested.	Reminders may be filtered.	Register for the production region; send test messages on all three networks.	Team	Before PH3
OI-05	No voice provider has been selected.	CX-1 cannot be met.	Evaluate providers behind the messaging adapter.	Team	PH3
OI-06	No trial supports a 2–hour second reminder.	An unevidenced default.	Second reminder off by default; measure before enabling.	Team	PH2
OI-10	Conflict rules for staff writes made offline need clinic input.	Data integrity at the desk.	Reject on conflict and show the staff member (FR-OFF-03); review in pilot.	Team	PH3
OI-11	Prior authorisation to process minors' sensitive data.	A launch gate for guardians.	Apply with the processing authorisation; the flow is built and demonstrable.	Clinic	Before PH3
OI-12	No statutory retention period for Cameroonian medical records was found.	Ten years retained, five shown is a proposal.	Confirm with counsel; values are tenant settings.	Clinic	Before PH3

CLOSED IN REV B	SUBJECT	CLOSED BY
OI-01	Routes lacked clinician and slot discovery; lifecycle lacked attendance states	FR-DIR-01, FR-DIR-02, FR-VIS-08
OI-07	Sample data used another country's conventions	AD-13, FR-TEN-06, FR-TEN-08
OI-08	An allergy field sat on the patient record	AD-06, FR-BKG-15, FR-PRV-10
OI-09	Ratings shown with no defined source	AD-11, FR-DIR-07 to FR-DIR-09

Amended 26 September 2026. NFR-SEC-08 puts AWS WAF in front of the web app and the API in production only. Development and test hold synthetic data, and a web ACL is a fixed monthly cost there with nothing to protect. CloudTrail stays in every environment.